

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 40%

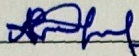
Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I Aparna.T (192524213) (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

SHORT ANSWERS TEST

COURSE: Cloud Computing

NAME: for big data Analytics

COURSE:

CODE: CSA1508

DATE: 21/07/2026

16/12

21/7/2026

- 1) Define cloud Computing and explain any four characteristics that make it different from traditional on-premise computing.
- 2) Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
- 3) Explain the role of server virtualization in cloud computing. Difference between virtualization as an enabling technology and cloud computing as a service model.
- 4) Discuss the role of data centers in delivering cloud services.
- 5) Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

14 DEFINITION:

* A Cloud Computing is a delivery ^{the} of computing services such as servers, storage, databases, networking, software and analytics using a Internet.

* It allows the users to access the demand without owning or maintaining any physical infrastructures.

CHARACTERISTICS:

1) On-demand self-services:

Users can access the computing resources through the physical servers and the networking based services.

2) Broad Network Access:

Services are available over the internet using the computing mobiles, laptops and tablets.

3) Reverse Pooling :-

Under the reverse pooling we may reverse the system through the cloud computing.

4> RAPID ELASTICITY :-

It may be contains the elasticity through the sufficient servers of and cloud computing.

3> ROLE OF SERVER VIRTUALIZATION :-

* SERVER Virtualization may divide the Physical virtual Address into multiple virtual Address (VM's). It will do one operation through the cloud computing.

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VIRTUALIZATION

It can differ from cloud computing and then it is an enabling technology.

CLOUD COMPUTING

It will differ from virtualization and it is a service model.

The virtualization

It can be used

may be used to store the remaining data which is not eligible in ram.

through the more companies like AWS, Google Cloud and then Microsoft.

The virtualization that may take place on all computers

The cloud computing will be decided through many software companies.

2) i) HARDWARE EVOLUTION:-

organization owned dedicated physical servers, which were expensive and underutilized.

ii) VIRTUALIZATION:-

Hypervisors allowed multiple virtual memory to run on one physical server, improving resource utilization.

iii) INTERNET EVOLUTION:-

It may contain the modular and underutilized Internet and broadband enabled through the remote access to computing servers.

iv) WEB APPLICATIONS:

Software may contains more reliable and presentable applications through the web applications.

v) CLOUD PLATFORM:-

Companies such as AWS, Microsoft Azure and Google cloud combined virtualization through this platform.

iv) ROLE OF DATA CENTERS IN DELIVERING CLOUD SERVICES:-

Data Centers store, manage server and network equipment.

Based on the user demand it increases and decreases.

It provide Internet very fast, Scalable to work in any where.

It also provide Backup System if any crash or issues occurs.

Enable the technology to upload the data.

3

Support Scalability by adding more resources when required.

5)

Feature	IaaS	PaaS	SaaS
Definition:	Provides virtual infrastructure such as servers, storage and networking.	Provides a platform to develop, test, and deploy applications.	Provides a ready-to-go software and over the Internet.
Real-world example	Build and manage IT infrastructure	Microsoft 365	Google.